Experiment 4-B

Vehicle Details Form Using Advanced Fields

Aim

To extend the Vehicle Details Form in Zoho Creator by incorporating advanced form fields to support vehicle verification, insurance document uploads, vehicle image handling, automatic vehicle identification, and additional vehicle information management in the Transport Management System.

Procedure

1. Open the existing Transport Management System application in Zoho Creator.
2. Navigate to the Vehicle Details Form and open the Form Builder.
3. Add advanced fields to extend the existing vehicle registration form.
4. Configure fields for vehicle image, insurance document upload, vehicle age, and additional vehicle information.
5. Use appropriate field types to store vehicle-related data such as currency, percentage, URL, and date-time wherever required.
6. Configure an Auto Number field to generate a unique Vehicle Code for each registered vehicle.
7. Add a Formula field if required to calculate vehicle-related values automatically.
8. Configure field validation rules to ensure correct data entry and restrict invalid values.
9. Save the form and enter sample vehicle details using the advanced fields.
10. Open the Vehicle Details Report and verify the stored records and uploaded documents.

Advanced Fields Used

Advanced Field	Field Name	Purpose
Decision Box	Vehicle_Availability	Indicates whether the vehicle is available for assignment.
Rich Text	Vehicle_Information_Display	Displays formatted vehicle-related instructions.
URL	Insurance_Policy_URL	Stores the vehicle insurance policy link.
Image	Vehicle_Image	Uploads the vehicle photograph.
Percent	Vehicle_Condition_Percentage	Records vehicle condition percentage, if assessed.
Currency	Vehicle_Maintenance_Cost	Stores vehicle maintenance expenses.
Decimal	Vehicle_Age_Years	Records vehicle age in years.
Date-Time	Vehicle_Registration_Timestamp	Stores the vehicle registration date and time.
SubForm	Vehicle_Maintenance_History	Maintains previous maintenance records.
Lookup	Assigned_Driver_Lookup	Selects a registered driver from Driver Details.

Signature	Vehicle_Inspection_Signature	Records the authorized vehicle inspection signature, if required.
File Upload	Insurance_Document_Upload	Uploads the vehicle insurance document.
Audio	Vehicle_Engine_Audio	Stores an optional engine sound recording.

Extended Form Layout

- **Section D, Vehicle Verification:** Assigned Driver Lookup, Vehicle Age, Vehicle Condition Percentage.
- **Section E, Vehicle Documents & Media:** Vehicle Image, Insurance Document Upload, Insurance Policy URL, Vehicle Inspection Video, Vehicle Engine Audio.
- **Section F, Maintenance Information:** Vehicle Maintenance History (SubForm), Vehicle Maintenance Cost.
- **Section G, Identification & Verification:** Vehicle ID, Vehicle Registration Timestamp, Vehicle QR Code, Vehicle Inspection Signature, Vehicle Information Display.

Validation & Configuration Rules

- Vehicle ID should be unique for every registered vehicle.
- Vehicle Age must contain a valid positive numeric value.
- Vehicle Maintenance Cost must accept non-negative values only.
- Vehicle Image should contain a valid image file.
- Insurance Document Upload should accept supported document formats.
- Assigned Driver Lookup should display registered drivers from the Driver Details Form.
- Vehicle Availability should be selected from the configured options.
- Formula fields should be read-only and calculated automatically when configured.
- QR Code generation should be configured using the vehicle identification value.
- Signature and media fields should be made mandatory only when required by the vehicle registration process.

Output Screenshots

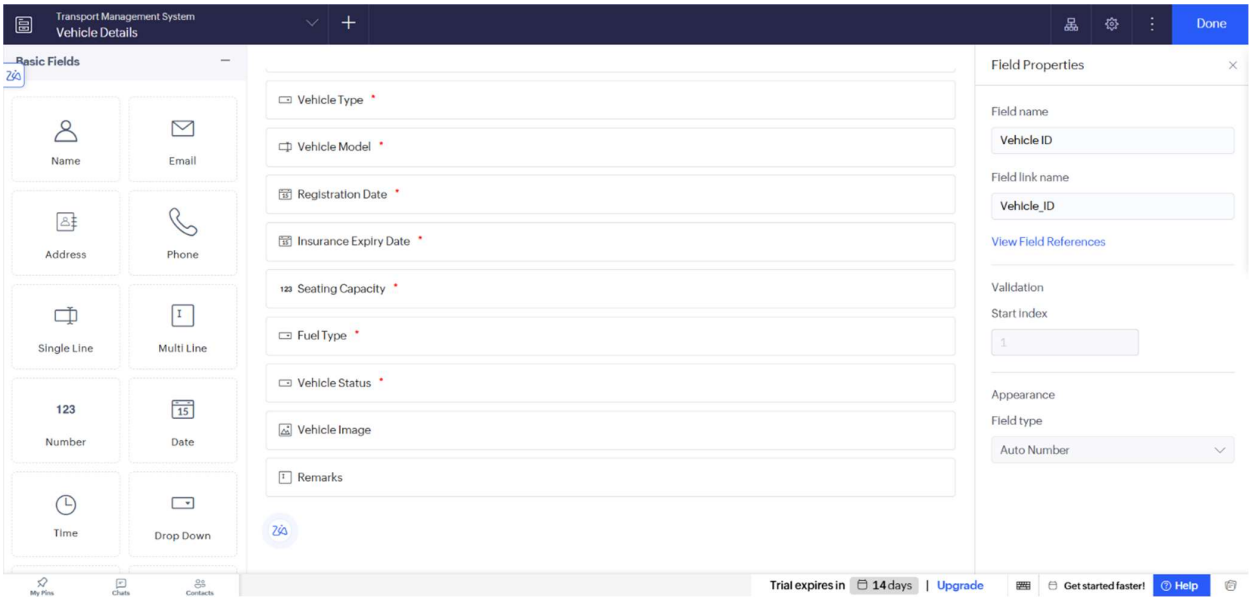


Figure 1: Vehicle Details Form Builder showing advanced fields such as Auto Number, Date-Time, lookup form.

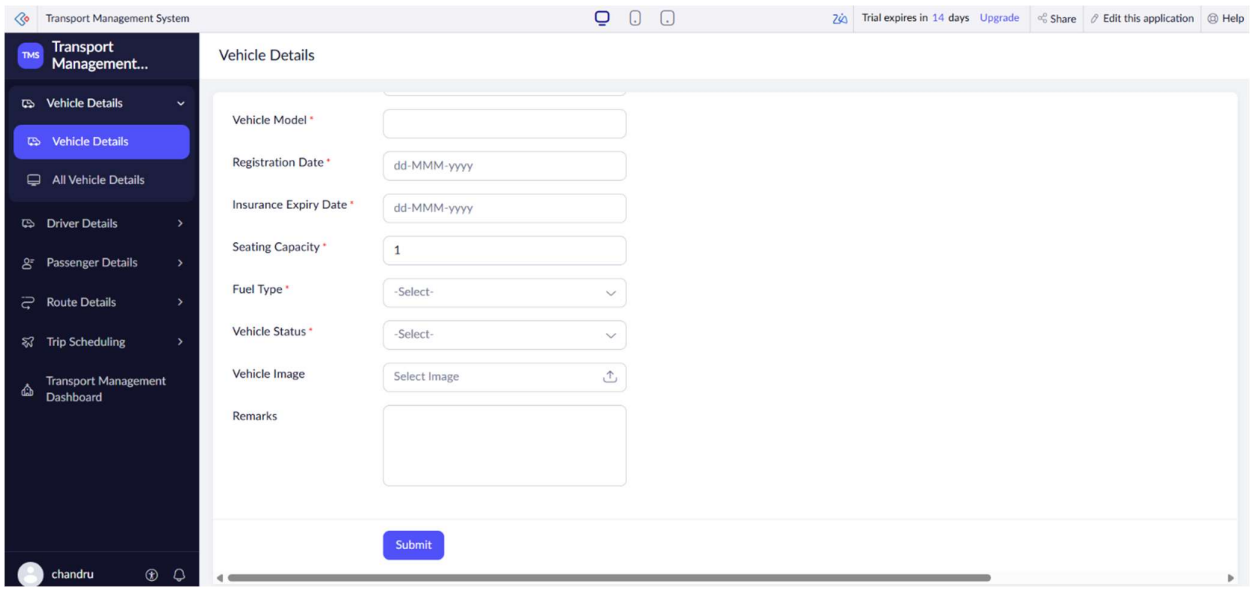


Figure 2: Live Vehicle Details Form displaying vehicle image, insurance document upload, driver lookup, and vehicle information.

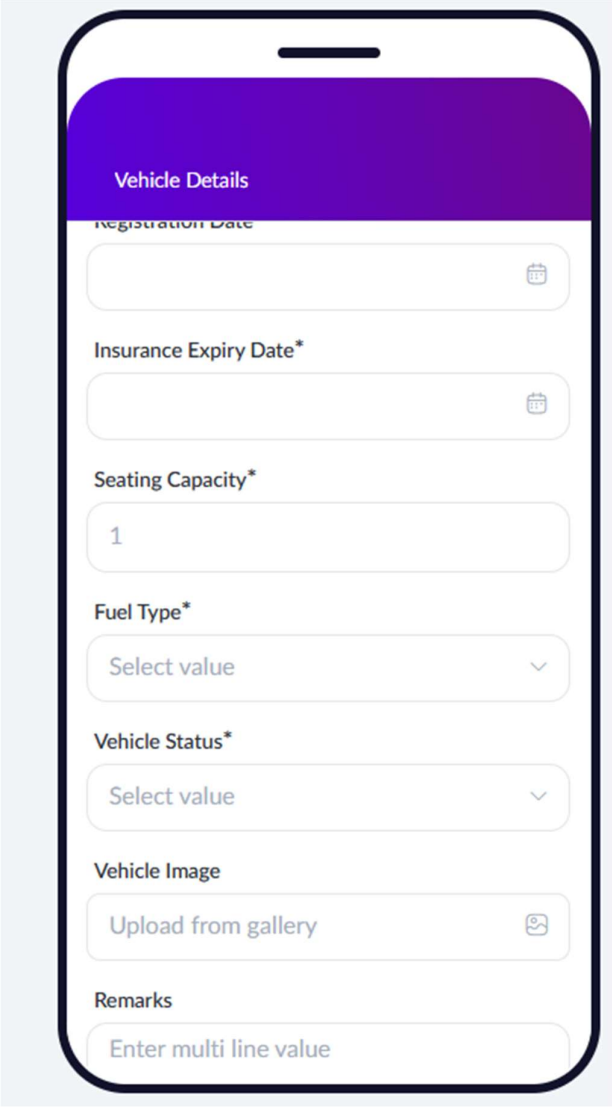


Figure 3: Vehicle Details Report displaying registered vehicle records with their identification codes and status.

Sample Record 1

Field	Value
Vehicle_ID	VEH-001 (illustrative)
Vehicle Number	TN 34 AB 5678
Vehicle Type	Car
Assigned Driver Lookup	Ramesh Kumar
Vehicle Age	3.5 Years
Vehicle Maintenance Cost	₹4,500
Vehicle Availability	Yes
Insurance Document	insurance_copy.pdf

Sample Record 2

Field	Value
Vehicle_ID	VEH-002 (illustrative)
Vehicle Number	TN 38 CD 1234
Vehicle Type	Bus
Assigned Driver Lookup	Arun Kumar
Vehicle Age	5 Years
Vehicle Maintenance Cost	₹8,000

Validation Test Cases

Field	Test Input	Expected Result
Vehicle Age	-5	Negative value should be rejected.
Maintenance Cost	-500	Negative currency value should be rejected.
Vehicle Image	Invalid image format	Unsupported image should be rejected.
Insurance Upload	Unsupported document format	File should be rejected.
Driver Lookup	Select registered driver	Rejected; maximum 45 seconds / 50 MB
Vehicle ID	Duplicate identification code	Duplicate should be prevented if uniqueness is configured.
Formula	Valid registration date	Vehicle age should be calculated if formula is configured.

Result

The Vehicle Details Form was extended using advanced fields in Zoho Creator to support vehicle identification, driver lookup, insurance document management, vehicle image uploads, and additional vehicle information. The advanced fields provide enhanced data collection and help improve vehicle record management in the Transport Management System.